Catholic Junior College
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME

CLASS 2T

PHYSICS 9478/03

Paper 3 Longer Structured Questions **15 September 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class in the boxes at the top of this page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

FOR EXAMINER'S USE	
SECTION A	
Q1	/ 11
Q2	/ 7
Q3	/ 9
Q4	/ 11
Q5	/ 5
Q6	/ 12
SECTION B	
Q7	/ 20
Q8	/ 20
PAPER 3	/ 75
PAPER 2	/ 75
PAPER 1	/ 30
PAPER 4	/ 50
TOTAL (WEIGHTED)	%

DATA:

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F$^{-1})$
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} mol^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

FORMULAE:

uniformly accelerated motion	s	=	$u\,t + \frac{1}{2}a\,t^2$
	v^2	=	$u^2 + 2\,a\,s$
work done on / by a gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / {}^{\circ}C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$
electric current	I	=	$nAvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R}$	=	$\frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C}$	=	$\frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$

energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0\, e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0\, e^{-\lambda t}$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this Section in the spaces provided.

1 **(a)** A positively charged metal sphere A is brought close to a negatively charged metal sphere B as shown in Figure 1.1. The charge on metal sphere B is twice that of the charge on metal sphere A.

On Figure 1.1, draw lines to represent the electric field around the spheres.



Figure 1.1 [2]

(b) Point P is at a distance *x* from the centre of sphere A along the line joining the centres of the two spheres as shown in Figure 1.2. The radius of A and B is 15 mm and the distance between the centres of the spheres is 80 mm.

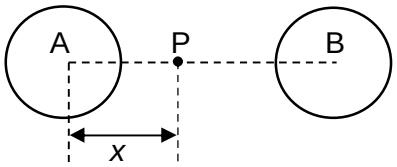


Figure 1.2

The variation with *x* of the electric potential *V* at P is shown in Figure 1.3.

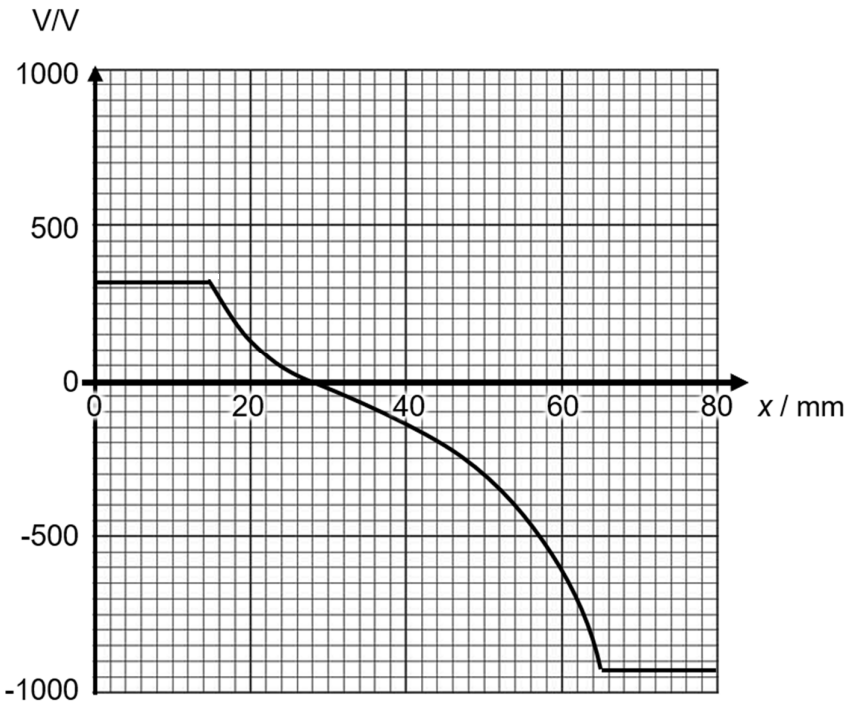


Figure 1.3

(i) Determine the magnitude of the electric field strength at P where x = 40 mm.

electric field strength = …….. V m^{-1} [2]

(ii) An electron is released from rest at point P where x = 40 mm.

1. Describe and explain the motion of the electron as it travels 20 mm along the line joining the centres of the spheres.

……..

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……..

……..

……..

……..

…… [3]

2. Determine the speed of the electron when it has travelled 20 mm along the line joining the centres of the spheres.

speed = ……..…….m s^{-1} [4]

[Total: 11]

2 On a computer motherboard where millions of operations can be performed every second, it uses a variety of capacitors for different operations. For example, 100 pF capacitors are used in high-speed areas to filter electrical noises.

(a) Figure 2.1 shows the variation of the charge stored Q on a 100 pF capacitor with potential difference V applied across the capacitor when charging the capacitor.

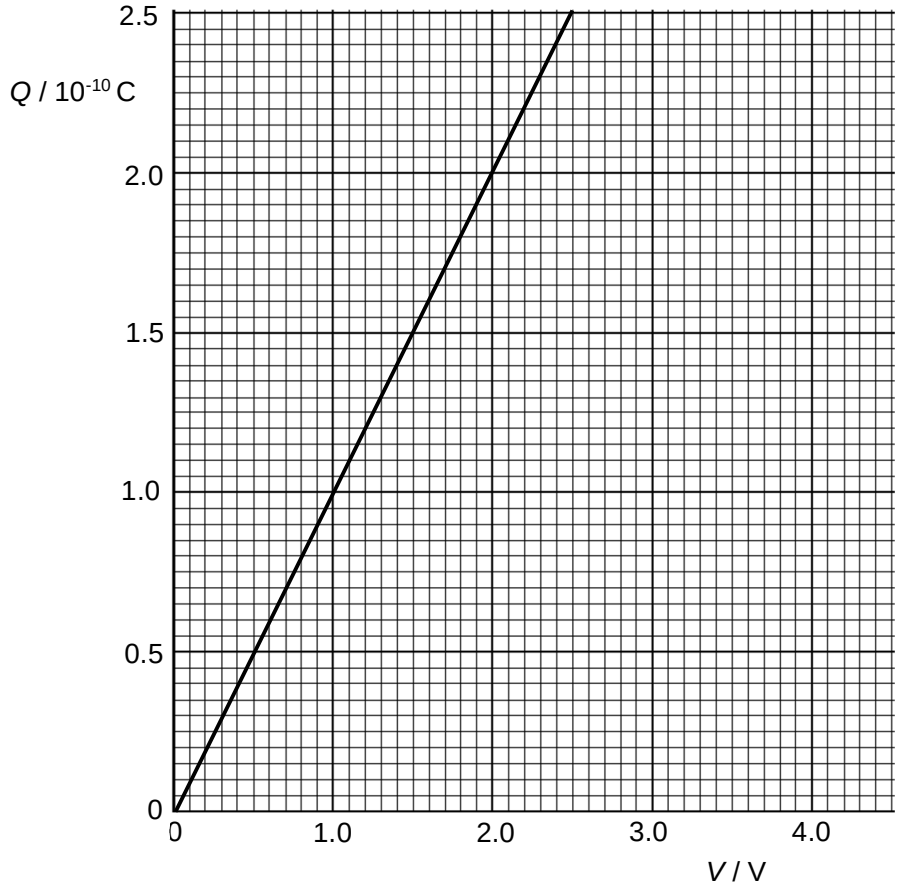


Figure 2.1

(i) Calculate the energy stored in the 100 pF capacitor when the potential difference across it is 2.5 V.

energy stored = ……..J [1]

(ii) While smaller capacitors are not able to store enough energy to meet sudden increase in current demands in circuitries, they are excellent at suppressing electrical noises.

On Figure 2.1, draw a line to show the variation of Q with V for a 25 pF capacitor. [1]

(b) Figure 2.2 shows six identical 1.0 μF capacitors that are connected in parallel. The capacitors are charged and discharged through the same 10 kΩ resistor. This circuit can be used on the computer motherboard to perform the power-up reset command.

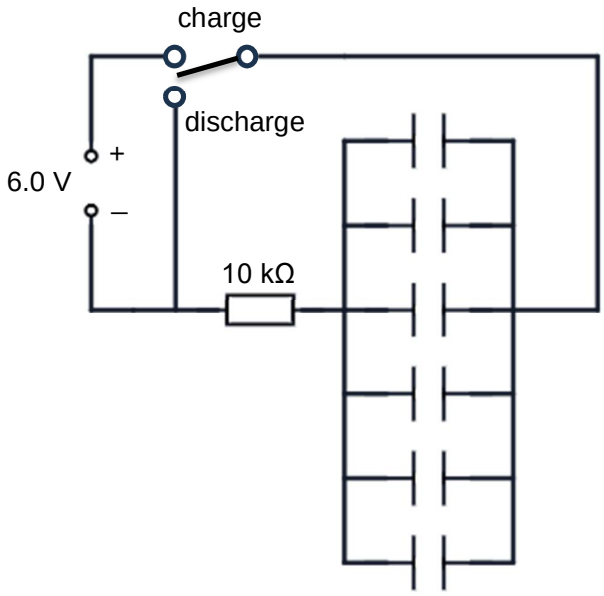


Figure 2.2

The power-up reset command can only be executed when the capacitors are fully charged to 6.0 V, which is equal to the voltage source. When the power-up reset command is executed, the capacitors start to discharge, and the command is completed when the capacitors discharge to 2.0 V or to approximately 33.3% of its maximum voltage V_o. You can take the maximum voltage V_o of the capacitors to be 6.0 V.

(i) Calculate the time taken to complete the power-up reset command.

time taken = ……s [2]

(ii) To protect the delicate circuits on the motherboard, the circuit is designed such that these capacitors can only start to charge again when its potential difference is equal to or less than 0.50 V. Thereafter, the capacitors charge again until they are ready for the next power-up reset command.

Determine the time taken for the capacitors to charge from 0.50 V to full charge. (Assume that a capacitor is fully charged when its potential difference is at 99.3% of its maximum voltage V_o.)

time taken = ……s [3]

[Total: 7]

3 **(a)** A solenoid shown in Figure 3.1 has length of 10 cm and 300 turns of wire. The current in the solenoid is 2.5 A when it is connected to a power supply.

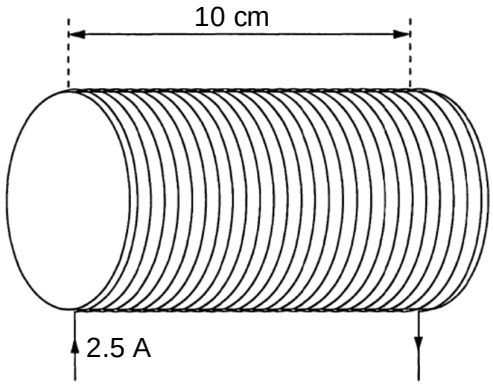


Figure 3.1

(i) Calculate the magnetic flux density inside the solenoid due to current of 2.5 A.

magnetic flux density = ……T [2]

(ii) Inside the dashed space provided in Figure 3.2, sketch the magnetic field pattern due to the current in the solenoid when 2.5 A of current is flowing in it as shown. Indicate the polarity of the magnetic field with **N** as the north pole and **S** as the south pole.

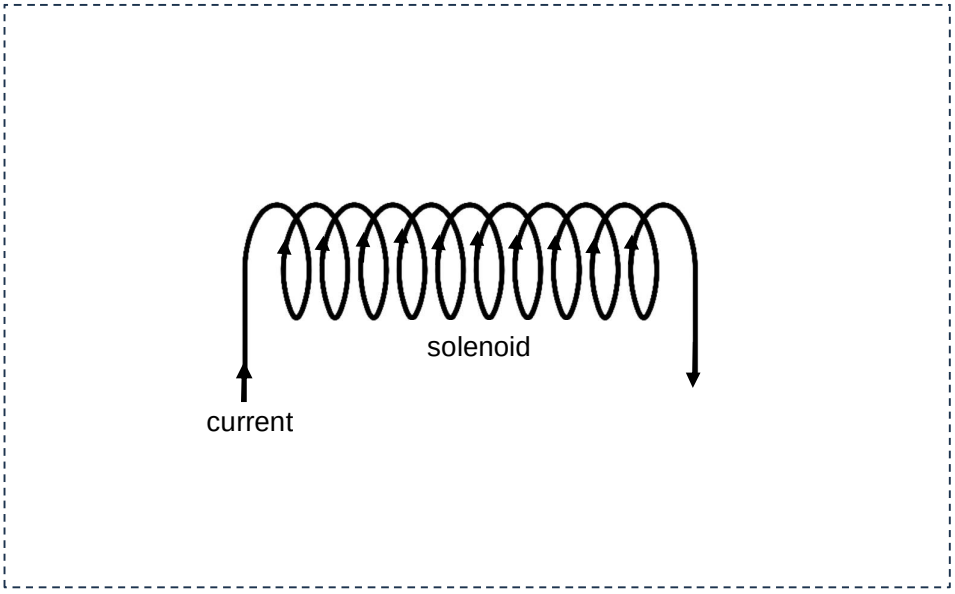


Figure 3.2 [2]

(b) A pair of concentric coils made of an inner coil Y placed within a solenoid is shown in Figure 3.3. The solenoid has cross-sectional area of 3.0 x 10^{-2} m^2 and is made of 2500 turns of wire. The solenoid is connected to a variable power supply by the terminals CD.

Coil Y has 500 turns of wire, a cross-sectional area of 7.5 x 10^{-3} m^2 and resistance of 3.0 Ω. Coil Y is connected to a resistor R of 15 Ω.

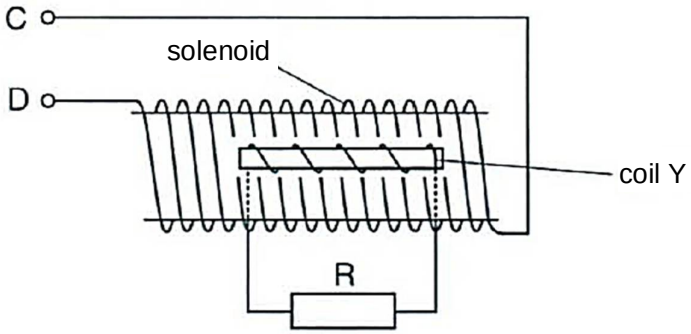


Figure 3.3

The current in the solenoid is then varied causing the magnetic flux density *B* in the solenoid to vary with time *t* as shown in Figure 3.4.

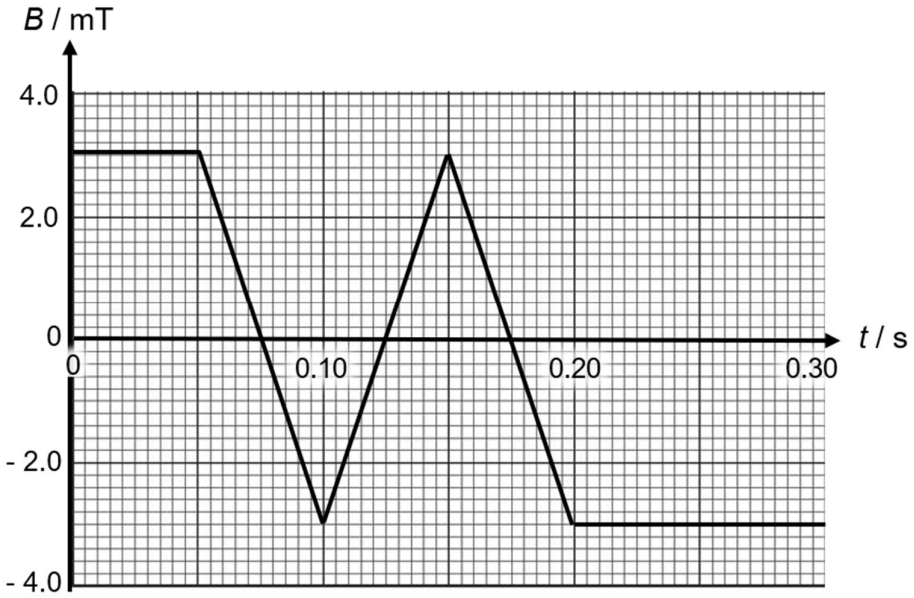


Figure 3.4

(i) Calculate the magnitude of the maximum current in R.

current = ……A [3]

(ii) On Figure 3.5, use your answer in **(b)(i)** to sketch a graph to show the variation with time of the current I in resistor R for time t = 0 to t = 0.30 s.

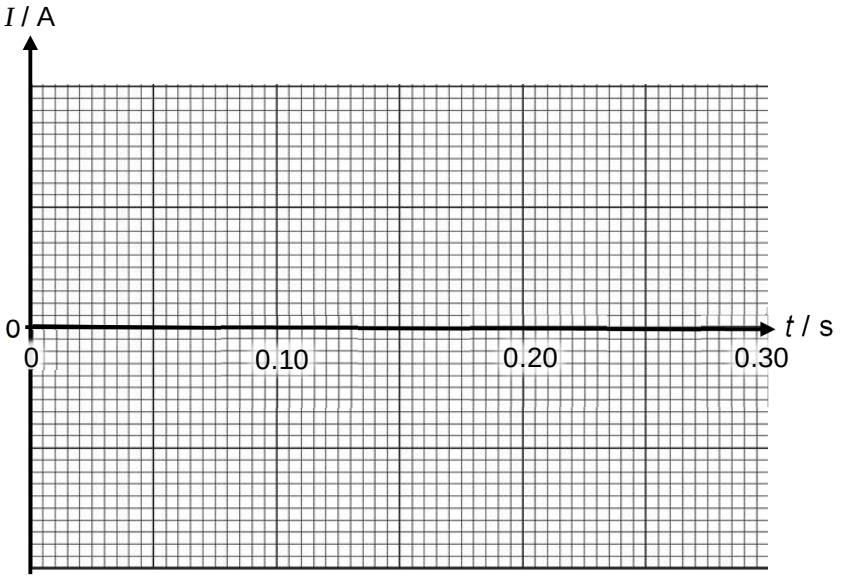


Figure 3.5 [2]

[Total: 9]

4 **(a)** Two light identical springs are joined in series and a 120 g bar magnet is hung from the lower end as shown in Figure 4.1. The effective spring constant of the combined springs is 11 N m^{-1}.

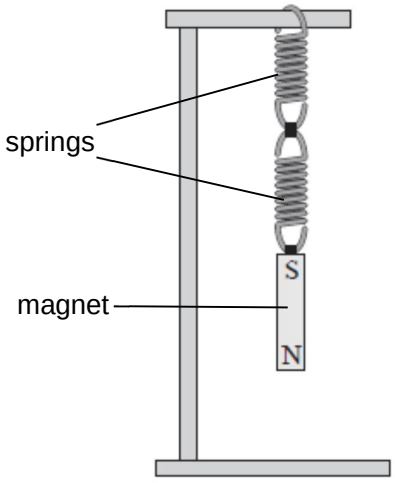


Figure 4.1

The bar magnet is displaced a small distance vertically downward from its equilibrium position and released. It then oscillates in simple harmonic motion.

(i) By considering the forces on the magnet at its equilibrium position and also at a small displacement x downward from its equilibrium position, show that, when released, the magnet's acceleration a is related to its displacement x by the equation:

$$a = -92x$$

Explain your working.

[3]

(ii) Using the expression in **(a)(i)**, calculate the angular frequency of oscillations of the magnet.

angular frequency = ……rad s^{-1} [2]

(iii) The amplitude of the oscillations of the magnet is 3.2 cm. The magnet first passes through the equilibrium position at time $t = 0$.

Calculate the magnitude of the displacement of the magnet from its equilibrium position when $t = 0.40$ s.

displacement = ……cm [2]

(b) Two identical systems shown in Figure 4.1 are now suspended with the North pole of their magnets located inside a coil of wire. The two coils are connected together with conducting wires as shown in Figure 4.2.

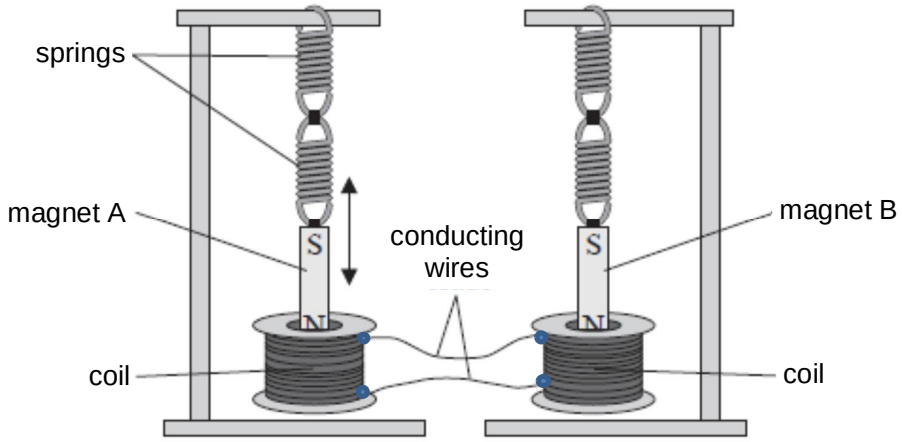


Figure 4.2

Magnet A is displaced so that it oscillates vertically. The North pole of magnet A moves into and out of the coil as it oscillates. As this motion continues, magnet B starts to oscillate. The amplitude of the oscillations of magnet B increases over time.

Explain why magnet B starts to oscillate and why the amplitude of oscillations increases.

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……... [4]

[Total: 11]

5 **(a)** Explain how line spectra, together with the concept of photon, provide evidence for discrete energy levels in atoms.

……

……

…… [2]

(b) A discharge lamp contains excited atoms which emit electromagnetic radiation at several wavelengths. Figure 5.1 shows the three lowest electron energy levels of one of these atoms, labeled n = 1, 2 and 3.

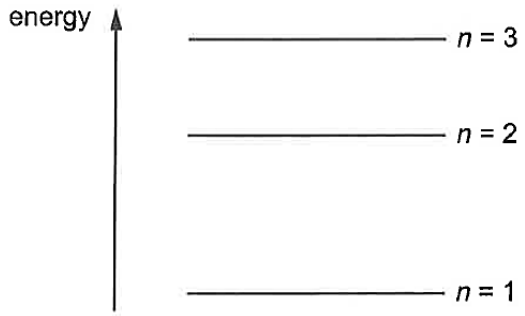


Figure 5.1

Electron transitions between these three energy levels can produce three different wavelengths of radiation. The transition from n = 2 to n = 1 produces radiation of wavelength 440 nm. Radiation of wavelength 590 nm is also emitted.

(i) Show that wavelength of the radiation emitted from the third transition is 250 nm.

[2]

(ii) State the region of the electromagnetic spectrum in which the third wavelength occurs.

…… [1]

[Total: 5]

6 **(a)** The isotope beryllium-7 $\left({}^{7}_{4}\mathrm{Be}\right)$ is radioactive with a half-life of 53 days. A beryllium-7 nucleus decays by the emission of a γ-ray photon of energy 0.48 MeV.

(i) Calculate the change in mass of a beryllium-7 nucleus when it undergoes γ-decay.

change in mass = ……kg [2]

(ii) Determine the probability of decay of a beryllium-7 nucleus in a time of 1.0 day.

probability = …….. [2]

(iii) Initially, the total mass of undecayed beryllium-7 in a radioactive sample is 5.7×10^{-12} kg.

Calculate the number of undecayed beryllium-7 nuclei in the sample after 120 days.

number = …….. [3]

(iv) Explain why, although the beryllium-7 is radioactive, the number of beryllium-7 nuclei in the sample does not change.

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…… [2]

(b) The beryllium-7 nucleus in **(a)** may be considered to be stationary before the emission of the γ-ray photon.

Determine the recoil speed of a beryllium-7 nucleus due to the emission of a γ-ray photon.

speed = ……m s^{-1} [3]

[Total: 12]

Section B

Answer **one** question in this Section in the spaces provided.

7 **(a)** State *Newton's second law of motion*.

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…… [2]

(b) A rollerblader approaches and glides up a smooth ramp of length 2.0 m along AB, inclined at an angle of 30˚ to the horizontal. The side view of this ramp is shown in Figure 7.1.

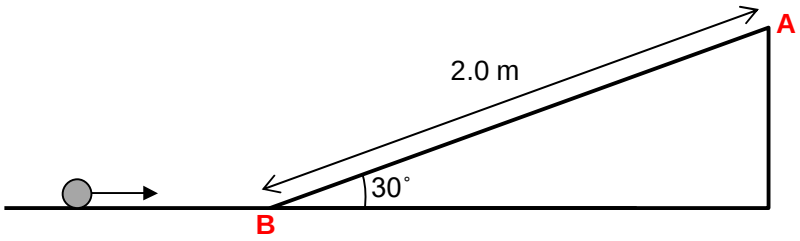


Figure 7.1

Figure 7.2 shows the ramp when viewed **perpendicular** to the ramp's inclined surface. The rollerblader glides up the ramp with initial speed u, at an angle of $\theta = 60^\circ$ to the base of the ramp as shown in Figure 7.2. He stops exerting energy the moment he starts moving up the ramp.

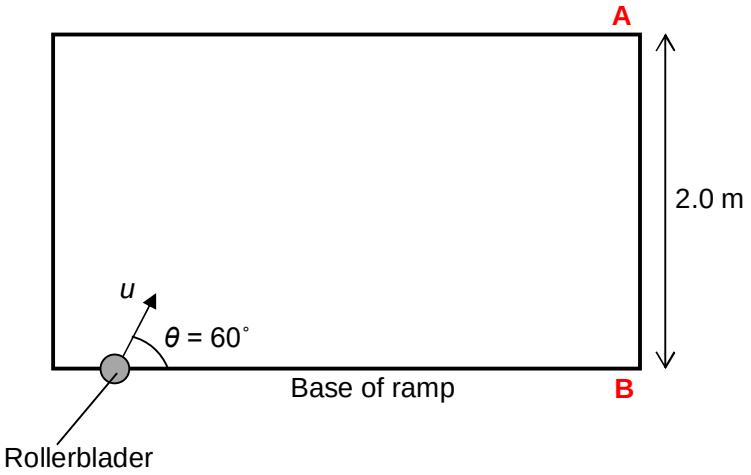


Figure 7.2

(i) Determine the magnitude of the acceleration of the rollerblader while moving up the ramp.

acceleration = ……..……m s^{-2} [2]

(ii) On Figure 7.2, draw an arrow to show the direction of the acceleration of the rollerblader. [1]

(iii) Explain why the rollerblader does not move up the ramp along a straight path but along a parabolic path.

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……... [3]

(iv) By using the principle of conservation of energy, determine the minimum speed *u* for the rollerblader to just make it to the top of the ramp.

minimum speed = ……..……m s^{-1} [3]

(v) Using the speed in **(b)(iv)**, determine the time taken for the rollerblader to return to the base of the ramp.

time taken = ……s [2]

(c) The ramp is extended to allow the rollerblader to move through a partial vertical circular motion with radius 3.0 m. Figure 7.3 shows the side view of the entire ramp. The rollerblader is upside-down at the top of the ramp.

The rollerblader now approaches the base of the ramp straight on perpendicularly.

You may regard the rollerblader as a point mass throughout this motion.

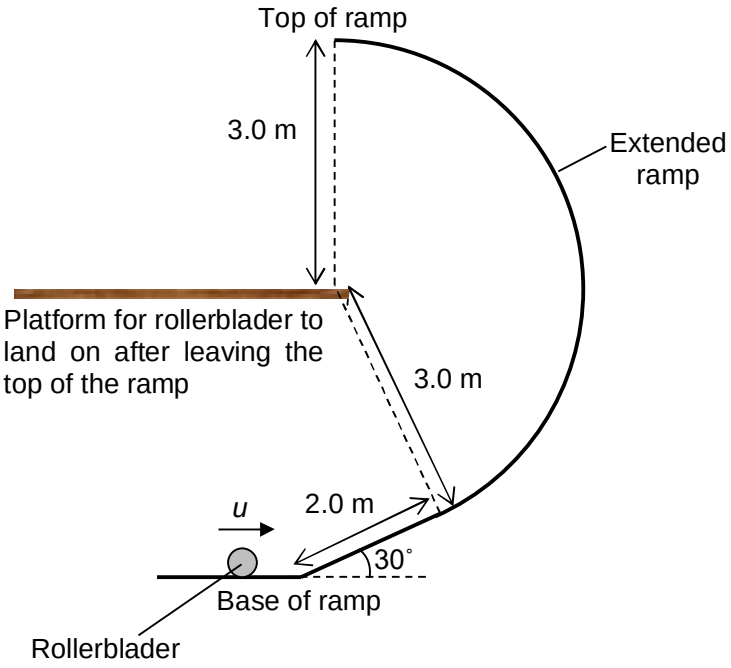


Figure 7.3 (not to scale)

(i) On Figure 7.4 below, sketch a graph to show how the magnitude of the centripetal acceleration a_c of the rollerblader varies with his linear speed v.



Figure 7.4 [1]

(ii) Determine the minimum speed u required for the rollerblader to reach the top of the circular ramp shown in Figure 7.3.

minimum speed = ……m s^{-1} [4]

(iii) On Figure 7.5 below, sketch a graph to show how the kinetic energy of the rollerblader E_K varies with his vertical displacement above the base of the ramp s_y when he moves up the ramp with this minimum speed.

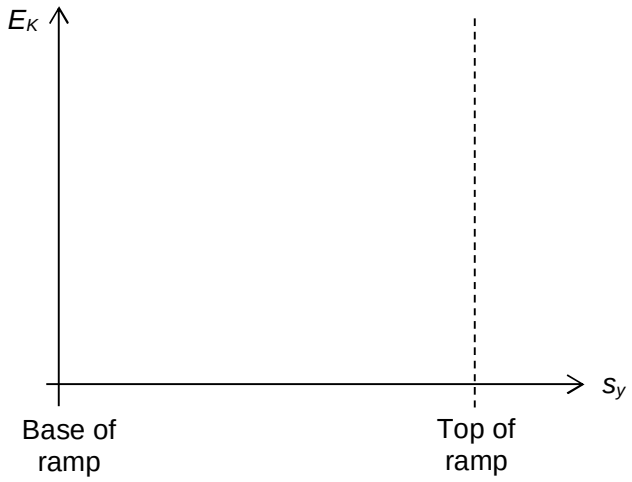


Figure 7.5 [2]

[Total: 20]

8 (a) (i) The Moon is continuously attracted towards the Earth by gravitational force and orbits around the Earth.

Explain why the Moon does not fall into the Earth.

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…… [2]

(ii) Assuming Earth to be a perfect sphere, explain why the weight of a person measured on an electronic balance at the poles is greater than that measured at the equator.

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…… [4]

(b) The Moon has an orbital period of about 27.3 days. The mass of the Earth is 6.00×10^{24} kg and the mass of the Moon is 7.34×10^{22} kg.

(i) Show that the distance between the centres of the Moon and the Earth is approximately 383 000 km.

[3]

(ii) Hence, calculate the distance *x*, from the centre of the Earth along the Earth-Moon line to the point Q where the resultant gravitational field strength due to the Earth and Moon is zero.

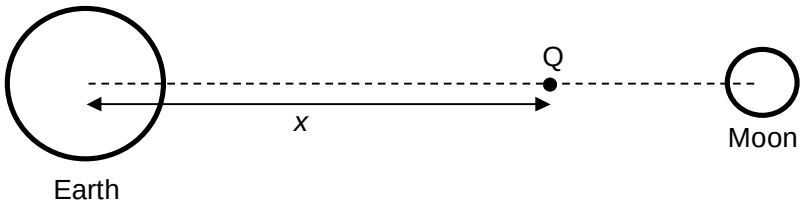


Figure 8.1 (not to scale)

x = ……..……..…….m [2]

(iii) Calculate the gravitational potential at point Q.

gravitational potential at point Q = ……..J kg^{-1} [2]

(iv) The radius of the Earth is 6370 km and the radius of the Moon is 1740 km.

Determine the minimum energy required to send a stationary capsule of mass 30 kg from the surface of the Earth to the surface of the Moon.

minimum energy = ……..J [4]

(c) A planet of radius *R* is isolated in space. It has no atmosphere, and its mass can be assumed to be concentrated at its centre.

A projectile is launched from the surface of the planet with a **minimum** kinetic energy ***E*** required for the projectile to **not** return to the planet.

On Figure 8.2, draw:

(i) a line, labelled **P**, to show how the gravitational potential energy of the projectile varies with distance from the centre of the planet as it moves from *R* to 4*R*,

(ii) a line, labelled **K**, to show how the kinetic energy of the projectile varies with distance from the centre of the planet as it moves from *R* to 4*R*.

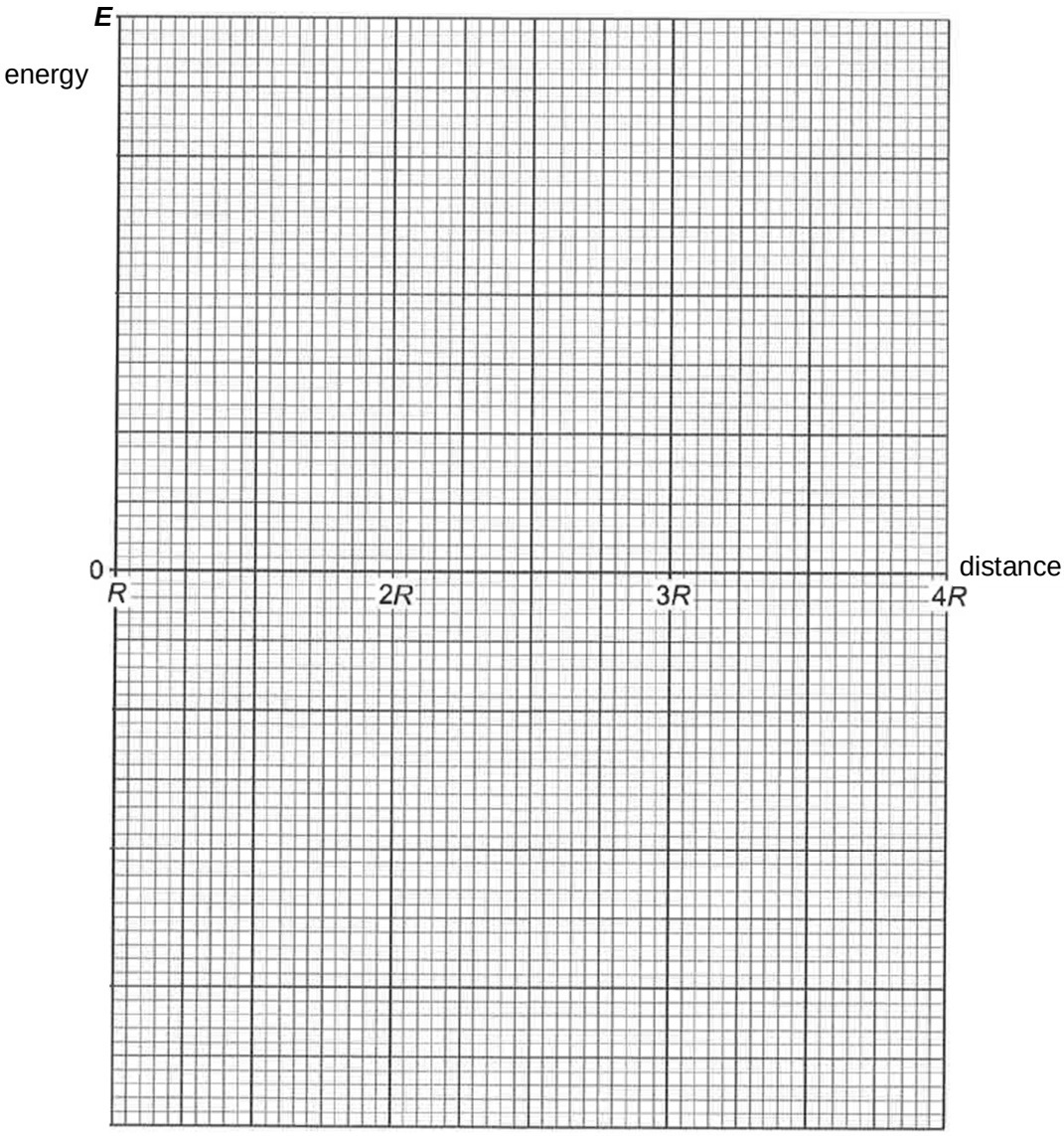


Figure 8.2 [3]

[Total: 20]

END OF PAPER 3

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